

User Authentication Module

The User Authentication Module is a core component of the Smart Car Recommendation System. It is responsible for managing user identity, ensuring secure access, and maintaining user-specific data such as preferences and saved cars. This module allows users to register (signup), login, and logout from the system. During registration, users provide basic details such as name, email, and password. The system validates the input and securely stores the data in the database. During login, the system verifies user credentials and grants access only if the details are correct. Authentication is handled using secure session management or token-based authentication (JWT), ensuring that only authorized users can access personalized features. The module also supports logout functionality, which safely ends the user session and prevents unauthorized access. Additionally, this module enables storing user-specific data such as wishlist, search history, and preferences. This helps in providing personalized recommendations and improving user experience. Technologies Used: - HTML: For creating user interface forms such as login and signup pages. - CSS: For styling the authentication pages and making them user-friendly and responsive. - JavaScript: For client-side validation and improving user interaction. - Python (Flask): For backend logic, handling authentication, session management, and API creation. - Database (MongoDB/MySQL): For securely storing user data and credentials. Overall, the User Authentication Module ensures secure, reliable, and personalized access to the system, forming the foundation for all user-related functionalities.